IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 24 June 2025, accepted 31 July 2025, date of publication 7 August 2025, date of current version 19 August 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3596643

RESEARCH ARTICLE

Masked and Noise-Masked Multimodal Brain Tumor Image Segmentation Using SegFormer and Shared Encoder Framework

K. HEMALATHA, P. R. VISHNU VARDHAN, ALFRED DHARMARAJ ARAVINDRAJ, AND S. HARI HARA SUDHAN

School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, Tamil Nadu 600127, India

Corresponding author: K. Hemalatha (hemalatha.kasirajan@vit.ac.in)

ABSTRACT Medical image segmentation is a critical task in clinical diagnosis and treatment, particularly for brain tumor analysis using imaging modalities such as Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) scans. However, segmentation is often hindered by noise introduced through scanner artifacts, motion blur, and inconsistent acquisition conditions. To address this limitation and the challenges of missing and noisy pixels in an MRI image, this work introduces the Masking and Noise-masking Multimodal SegFormer (MNMS) - a transformer-based framework designed for brain tumor segmentation, which collectively combines complementary information from multiple MRI modalities while integrating dual masking strategies, one to handle incomplete data and another to overcome noise artifacts. Being a multimodal framework, MNMS can effectively work with and provide valuable segmentation results for any single modality which it has been trained for, ensuring robustness in real-world clinical scenarios where multimodal data may not always be available. MNMS employs masking to overcome computational complexity while maintaining local and global features essential for accurate medical image segmentation. Additionally, noise-masking introduces a controlled Gaussian noise into the MRI images creating random variations in the pixel intensities and thereby encouraging the model to learn the invariant and essential patterns in MRI images. Evaluation on Brain Tumor Segmentation (BraTS2020) challenge dataset demonstrates that MNMS outperforms conventional convolutional neural network (CNN) based methods, achieving superior accuracy and Dice Similarity Coefficient (DSC) scores. Specifically, the proposed MNMS model achieved a DSC score of 0.9341, outperforming UNEt TRansformers (UNETR), which has a DSC of 0.8208 and 3D U-Net (0.8179). These results highlight its effectiveness in multimodal brain tumor segmentation, ultimately contributing to improved diagnostic accuracy and patient care.

INDEX TERMS Medical image segmentation, transformers, SegFormer, masking and noise-masking, multimodal images, feature extraction.

I. INTRODUCTION

Medical image segmentation plays a crucial role in disease diagnosis, treatment planning, and prognosis by enabling precise identification of anatomical structures and abnormalities in medical images. This process assists healthcare professionals in making informed decisions, ultimately leading to better patient outcomes. However, achieving high-accuracy segmentation remains a significant challenge due to the complex nature of medical images, variations in imaging modalities, and the presence of missing or corrupted data. Corrupted data can arise due to noise, artifacts, low resolution, motion blur, or inconsistencies introduced during image acquisition, compression, or transmission. This can deeply affect the segmentation quality, making it difficult for the models to extract meaningful features. In addition, existing segmentation models often face challenges with missing modalities - a common problem in real-world

The associate editor coordinating the review of this manuscript and approving it for publication was Ayman El-Baz.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

clinical settings where complete multimodal imaging data isn't always accessible. The incompetence of the models to efficiently use incomplete or degraded data hinders the flexibility and feasibility of these models, stressing the need for solutions that can tackle these issues effectively.

While CNNs [1], [2], [3] have shown potential in medical image segmentation, their small receptive fields and emphasis on local features make it challenging for them to capture the bigger picture. This becomes even more difficult in multimodal scenarios, where integrating information from multiple imaging modalities is crucial for accurate segmentation. Therefore, improving the segmentation models to tackle missing or corrupted data while understanding broader contextual information is vital to improve their real-world applicability.

To overcome the challenge faced in multimodal data, transformer-based architectures have been introduced in medical image segmentation [4], providing an in depth understanding of global context and feature representation. Unlike CNNs, transformers use self-attention mechanisms, allowing them to capture long-range dependencies and correlations within images such as spatial layouts, structural patterns, and contextual links between different regions. This helps in precisely differentiating anatomical structures, identifying abnormalities, and ensuring accurate segmentation by considering the image as a whole, rather than focusing primarily on localized features. As a result, transformers are ideal for complex medical segmentation tasks.

One notable model, SegFormer, has showcased both efficiency and competitive performance in 2D image segmentation by integrating hierarchical feature extraction with lightweight computation, reducing the need for extensive pretraining. Moreover, combining noise [5] into the training process restricts the models from over-relying on certain features, improving better generalization. These strategies help transformers focus on the most relevant and unique features within medical data, improving segmentation accuracy and ensuring a more promising performance, even when the data is incomplete or noisy.

Building upon these advancements, our proposed framework introduces a robust solution to the limitations posed by incomplete and noisy medical imaging data through the integration of both window-masking and noise-masking strategies within a multimodal transformer-based architecture. By selectively obscuring parts of the input (window-masking), the model is trained to infer missing regions using surrounding context, enhancing its resilience to incomplete inputs. Simultaneously, noise-masking simulates real-world distortions by introducing controlled Gaussian noise, encouraging the model to learn invariant and robust features. This dual-masking approach ensures that the model focuses on essential tumor patterns while remaining insensitive to irrelevant perturbations. When fused with modality-specific and shared encoders, the framework effectively captures both distinct and complementary features from various MRI sequences. Together with the SegFormer backbone, which leverages self-attention for global context understanding, our approach offers a resilient and generalizable solution for accurate brain tumor segmentation, even in the presence of corrupted or partially missing data.

Unlike conventional CNN-based segmentation models like U-Net and 3D U-Net, which often use localized receptive fields and have trouble capturing global contextual relationships, the MNMS framework uses a transformer-based architecture that can model long-range dependencies in MRI data. Traditional models are particularly vulnerable to performance degradation in the presence of missing or corrupted input, as they lack explicit mechanisms to simulate or adapt to such inconsistencies. MNMS overcomes this by using its dual-masking strategy, which actively trains the model to infer missing regions and ignore noise-induced distortions. Moreover, instead of shallow feature concatenation, MNMS utilizes a modality-specific and shared encoder fusion mechanism, allowing both distinct and complementary information from T1, T2, and FLAIR images to be effectively captured and integrated. This fusion strategy, combined with hierarchical attention from SegFormer, enables the model to generalize better across heterogeneous MRI datasets, making it a more robust and practical solution for complex brain tumor segmentation tasks.

The main contributions of this work can be summarized as follows:

1) To address the challenges of missing and noisy pixels in an MRI image, a Masking and Noise-Masking Multi-modal Segformer (MNMS) model is proposed which collectively combines complementary information from multiple MRI modalities while integrating dual masking strategies, one to handle incomplete data and another to overcome noise artifacts.
2) In this framework window-masking is implemented into the images of different modalities to overcome computational complexity while maintaining local and global features essential for accurate medical image segmentation.
3) To address the challenges of missing pixels and noisy MRI images, a noise-masking strategy is proposed, which introduces a controlled Gaussian noise into the MRI images creating random variations in the pixel intensities and thereby encouraging the model to learn the invariant and essential patterns in MRI images.
4) In this work, we employ a specific and shared encoder framework to effectively handle various multi-modal images, establishing balance between modality specific feature extraction and cross modal feature learning. This approach enhances the segmentation accuracy and handles missing modalities efficiently.
5) To the best of our knowledge, this is the first ever work which proposes to integrate both masking and noise-masking strategies in a single multi-modal brain tumor segmentation framework. By strategically masking and variating pixel intensities, the model tunes

itself to learn the invariant and essential patterns in multi-modal MRI images.

The remainder of this paper is structured as follows: Section II reviews related work in medical image segmentation and hybrid modeling approaches. Section III details the proposed methodology, including preprocessing, model architecture, and learning strategies. Section IV presents the experimental setup, results, and evaluation metrics. Finally, Section V concludes the study and outlines potential future research directions.

II. RELATED WORK

In medical image segmentation, challenges like capturing fine-grained spatial information from confined training data and improving feature representation have been solved due to significant improvements with the rise of deep learning (DL) approaches. Techniques such as noise-masking and contrastive learning have further enhanced feature extraction, making models more adaptable to variations in medical imaging data. This section briefs about the key improvements in medical image segmentation, the use of DL models, and latest advancements in multimodal and noise-masking techniques.

A. MEDICAL IMAGE SEGMENTATION

MT-UNet, developed by Wang et al. [6] integrates a Mixed Transformer Module (MTM) using Local-Global Gaussian-Weighted Self-Attention (LGG-SA) for intra-affinity learning and External Attention (EA) for inter-sample correlation. It outperforms CNNs in accuracy, models long-range dependencies, and reduces computational complexity across various medical image segmentation datasets.

Mini-Net, introduced by Javed et al. [7] has under 38,000 parameters, enabling efficient segmentation in real-time applications. Despite its compact size, it captures both high- and low-frequency features effectively and performs well on datasets like DRIVE, STARE, ISIC-2016/2018, and MoNuSeg.

Res-PAC-UNet by Ansari et al. [8] segments liver CT with 1.2M parameters using a residual backbone and Pyramid Atrous Convolution (PAC) for multiscale extraction. Achieving a dice score of 0.958 on the MSD liver dataset, it combines low overhead with high accuracy, suitable for constrained clinical environments.

B. TRANSFORMER-BASED MODELS

Collaborative Multi-Head Attention (CMHA), proposed by Cordonnier et al. [9] optimizes multi-head attention by reducing computational costs without sacrificing performance. This re-parameterized transformer is efficient for natural language processing (NLP), neural machine translation (NMT), and computer vision tasks.

MAViT-UNet, developed by Yu et al. [10] combines U-Net with Vision Transformers (ViT) and Multi-Head Attention to enhance medical image segmentation. It leverages transformers for global context and U-Net's skip connections for local feature retention, improving accuracy on limited-label datasets.

C. MULTIMODAL TECHNIQUES

MDU-CNN, introduced by Srinivasan et al. [11] improves multimodal biomedical image segmentation by integrating multi-dimensional convolutional layers into U-Net, enhancing spatial feature extraction across MRI, dermoscopy, and EM datasets.

DRIFA-Net by Dhar et al. [12] uses Multi-branch Fusion Attention (MFA) and Multimodal Information Fusion Attention (MIFA) for dual-attention fusion, improving classification for MRI, CT, and dermoscopy. Monte Carlo dropout adds uncertainty estimation, boosting model reliability.

Ansari et al. [13] reviewed deep learning for ultrasound segmentation, highlighting transformer-based models, Fully Convolutional Networks (FCN), U-Net variants, attention mechanisms, and Generative Adversarial Networks (GAN) based data generation. It addresses issues like speckle noise, low contrast, and organ-specific challenges.

Another review by Ansari et al. [14] summarizes GAN-based EUS generation from B-mode ultrasound for breast cancer diagnosis. A U-Net generator with tumor-aware discriminator achieves Structural Similarity Index (SSIM) of 0.903 on multicenter data, aiding handheld devices in low-resource settings. The review also stresses the need for validation diversity and interpretability for clinical safety.

Recent work by Dörterler et al. [15] explores the hybridization of meta-heuristic algorithms with K-means for medical data clustering, showing improved convergence speed and cluster accuracy. In a related study, Dörterler et al. [15] combined K-means with Particle Swarm Optimization and Ant Colony Optimization to enhance heart disease diagnosis, optimizing clustering and classification phases. These approaches highlight the relevance of hybrid ML models in medical decision support systems, especially for multimodal patient data analysis.

D. MASKING AND NOISE MASKING TECHNIQUES

Denoising Self-Distillation Masked Autoencoder (DSDMA), proposed by Xu et al. [16] combines masked autoencoders (MAE) with a teacher-student framework to enhance feature extraction and robustness to noisy medical data. It outperforms contrastive methods on CIFAR-10, CIFAR-100, and STL-10.

To improve semantic representation, Xie et al. [17] introduced MedIM, which uses radiological reports for knowledge-driven (KDM) and sentence-driven masking (SDM). MedIM surpasses traditional Masked Image Modeling (MIM) in classification and segmentation tasks.

E. CROSS-DOMAIN DEEP LEARNING APPLICATIONS FOR SCIENTIFIC TIME-SERIES AND IMAGING DATA

Recent advances in biomedical signal processing showcase deep learning's potential beyond imaging. Afsa et al. [27]

TABLE 1. Summary of recent methods in medical image segmentation with their challenges, techniques, and outcomes.

Citation	Problem	Methodology	Dataset	Remarks
Ronneberger et al., (U-Net) [18]	Limited annotated data for cell segmentation	U-Net with weighted loss and elastic deformation	PhC-U373, DIC-HeLa, EM Drosophila	Introduced U-Net; enabled precise segmentation with few annotations
Wang et al., [19]	Difficulty in tumor segmentation and survival prediction	3D U-Net + VAE branch for regularization	BraTS 2019 (HGG, LGG)	Combined segmentation and survival prediction with moderate accuracy
Zhao et al., (MMGL) [20]	Limited labeled data in cardiac segmentation	Multi-scale, multi-view contrastive learning	MM-WHS (CT volumes)	Outperformed contrastive baselines; improved local-global feature capture
Zhou et al., (DEU-Net) [21]	Pixel-wise methods struggle in 3D segmentation	Dual encoder: Transformer + CNN with DFFM	BraTS 2020	Enhanced contextual feature learning; strong segmentation accuracy
Huang et al., (DRA-Net) [22]	CNN/Transformer struggle with local details	Regional attention + shift MLP for edge detail	GLAS, CoCaHiE	Improved segmentation via edge-aware modules; effective feature fusion
Perera et al., (Seg-Former3D) [23]	Large models are resource-heavy, slow	Lightweight transformer with MLP decoder	BraTS, Synapse, ACDC	33× fewer parameters; efficient and high-performing segmentation
Wang et al., (ShaSpec) [24]	Handling missing modalities in multi-modal learning	Shared-specific encoder framework	BraTS2018, AudioMNIST	Strong performance under missing modality conditions
Xu et al., [25]	Poor abstraction in self-supervised models	Denoising self-distilled MAE	CIFAR-10/100, STL-10	Effective self-supervised learning; enhanced fine-tuning capability
Wang et al., (MPS-AMS) [26]	Limitations in current masked image modeling	Adaptive masking + patch selection	BUSI, Hecktorr, BraTS2018	Reduced gradient noise; improved lesion-focused representation

proposed the StrataChron Pyramid Fusion Framework (SPFF) for early cardiac arrest prediction from multivariate time-series and EHR using multi-scale temporal fusion and ICA. SPFF achieves high interpretability and robustness to class imbalance, with an AUROC of 0.998 and recall of 0.989 on the MIMIC-IV dataset. It predicts cardiac arrest up to 15 hours in advance and generalizes to other critical conditions using SHAP and ICE tools.

Ansari et al. [28] compared AttiaNet and ResNet1D for ECG-based Heart Age Estimation across signal lengths and sampling rates, finding improved robustness and efficiency in low-resource settings via data augmentation and transfer learning.

Though distinct from biomedical use, geoscientific image analysis also benefits from lightweight, interpretable CNNs. FluidNet-Lite, proposed by Yaqoob et al. [29] models two-phase flow in heterogeneous porous media as pixel-wise classification using a Physics Embedding Module and Grain-Weighted Adaptive Loss. Achieving high IoU (0.92) and SSIM (0.89), it is 94% lighter than GANs while remaining robust across flow regimes.

A cross-dataset evaluation of deep learning models for thyroid cancer recurrence was conducted by Dursun et al. [30] integrating clinical and histopathological features. The study emphasizes the need for generalizable models across institutions by addressing distribution shifts in medical data. Results highlight key risk factors and validate the model's robustness across varying dataset sources, improving recurrence prediction.

F. RESEARCH GAPS AND CONTRIBUTIONS

Despite notable progress in medical image segmentation using deep learning, several challenges continue to limit the clinical reliability and generalization of existing models. Many approaches rely heavily on single-modality imaging, which lacks the diversity needed to represent complex tumor structures such as enhancing regions, edema, and necrosis. This restricts the model's ability to adapt to varied clinical scenarios. Additionally, conventional CNN-based architectures often struggle to capture long-range spatial dependencies and are sensitive to noise, motion artifacts, and missing data - conditions that are common in real-world medical environments. Existing multimodal approaches frequently apply shallow or naive fusion techniques that fail to balance modality-specific and shared features effectively, leading to suboptimal segmentation outcomes. Furthermore, few models are designed to learn robust representations in the presence of corrupted or incomplete inputs, which hinders their resilience and clinical applicability. Addressing these gaps is essential for developing segmentation models that are robust, adaptable, and clinically viable.

III. METHODOLOGY

A. OVERVIEW

The proposed MNMS framework begins with a comprehensive preprocessing pipeline applied to the MRI modalities - T1, T2, and FLAIR. Each modality is converted from Neuroimaging Informatics Technology Initiative (NIfTI) format to Portable Network Graphics (PNG) format and standardized in terms of size (256, 256, 3) and intensity [0,1] to ensure uniform input. Dual masking strategies are then applied independently to each modality: window-masking randomly removes patches to simulate missing data, while noise-masking adds controlled Gaussian noise to imitate real-world acquisition artifacts. These masked versions help the model become resilient to imperfections in the data. The masked images are then passed through modality-specific encoders that extract unique feature maps capturing anatomical structures and modality-specific data. Notably, the model is designed to be flexible - it can perform

effectively even when only one MRI modality is provided, ensuring robust segmentation in scenarios where multimodal data may be incomplete or unavailable.

The extracted features from each modality are then processed through shared encoders that merge and refine the feature representations into a unified latent space. These encoders operate on both noise-masked and window-masked feature streams, ensuring that the model learns to infer missing patterns while disregarding noisy artifacts. The fused representations are then aligned using a transformation module and passed into the core MNMS segmentation backbone. Here, a transformer-based encoder processes the inputs using hierarchical multi-scale attention to capture both local details and global spatial dependencies. A lightweight multilayer perceptron (MLP) based decoder then reconstructs high-resolution segmentation masks, which are further refined using softmax to assign class probabilities to each pixel. This integrated design, combining modality-specific encoding, cross-modal fusion, and adaptive masking, enables MNMS to deliver precise tumor segmentation.

The architecture pipeline of the proposed framework is shown in Figure 1.

B. DATASET PREPARATION AND PREPROCESSING

1) DATASET DESCRIPTION

The BraTS 2020 dataset is a collection of multimodal MRI images specifically curated for brain tumor segmentation tasks. It is part of the annual Brain Tumor Segmentation (BraTS) challenge, held during the Medical Image Computing and Computer-Assisted Intervention (MICCAI) conference since 2012. The dataset provides high-quality, annotated MRI scans, serving as a benchmark for developing and comparing algorithms for automatic tumor segmentation and classification.

The BraTS 2020 dataset includes MRI scans from patients with brain tumors, containing both healthy tissue and tumor regions, with detailed annotations for training and evaluation purposes. The goal is to segment different tumor regions, as outlined in Table 2.

Each subject's data consists of four MRI modalities:

TABLE 2. MRI modalities used in the dataset.

MRI Modality	Description
T1-weighted (T1)	Provides high-resolution anatomical images for structural assessment.
T1-weighted post-contrast (T1ce)	Captured after contrast agent injection, enhancing tumor visibility.
T2-weighted (T2)	Improves contrast between different tissue types, aiding in tumor boundary detection.
Fluid-attenuated inversion recovery (FLAIR)	Highlights edema (swelling) around tumors by suppressing fluid signals, crucial for segmentation tasks.

The dataset is separated into three subsets for testing, validation, and training in order to assure accurate model evaluation. In order for the model to learn and capture both local and global features from multimodal inputs, the training set is essential.

TABLE 3. Distribution of MRI Modalities across Training, Validation, and Testing Sets.

Modality	Training Set	Validation Set	Testing Set	Total Images
T1	1255	275	265	1795
T2	1255	275	265	1795
FLAIR	1255	275	265	1795
Total	3765	825	795	5385

By testing the model's performance on unknown data, the validation set is essential for adjusting hyperparameters to minimise overfitting.

Lastly, the test set offers a fair evaluation of the model's segmentation accuracy, assuring that the outcomes matches actual clinical uses. Table 3 displays the BraTS 2020 dataset's overall distribution, which includes the balanced distribution across modalities and the structured division into training, validation, and test sets. This well-organised setup is essential for confirming the accuracy and generalisability of the proposed multimodal segmentation strategy.

The segmentation tasks in this work's studies are carried out using the T1, T2, and FLAIR modalities.

2) DATA PREPROCESSING

In the pre-processing step for multimodal brain tumor image segmentation, necessary measures are taken to prepare the raw data from T1, T2, and FLAIR modalities for effective training. The first step is to convert NIfTI (.nii) files to PNG format. The NIfTI is an open file format that is used to save data related to brain imaging that is captured using MRI.

Most deep learning libraries are built to handle standard image formats like PNG, Joint Photographic Experts Group (JPG), or Bitmap Image (BMP). NIfTI files store 3D volumetric data and sometimes time series data - which is more complex to work with. Converting to PNG format allows conversion of the 3D volume into 2D slices like axial, coronal, or sagittal views - making them compatible with transformer-based models.

Another crucial step involves normalizing the intensity values of each modality to a standardized range of [0, 1]. This normalization ensures that each modality has comparable intensity distributions, which aids in stable model training. The normalization equation is:

$$I_{\text{norm}} = \frac{I - \mu}{\sigma} \tag{1}$$

where I is the raw image, μ is the mean, and σ is the standard deviation of pixel intensities.

Alongside normalization, various standard pre-processing techniques are applied. All images from the T1, T2, and FLAIR modalities are resized to a consistent input dimension (H, W, C), typically $H = 256$ and $W = 256$, to maintain

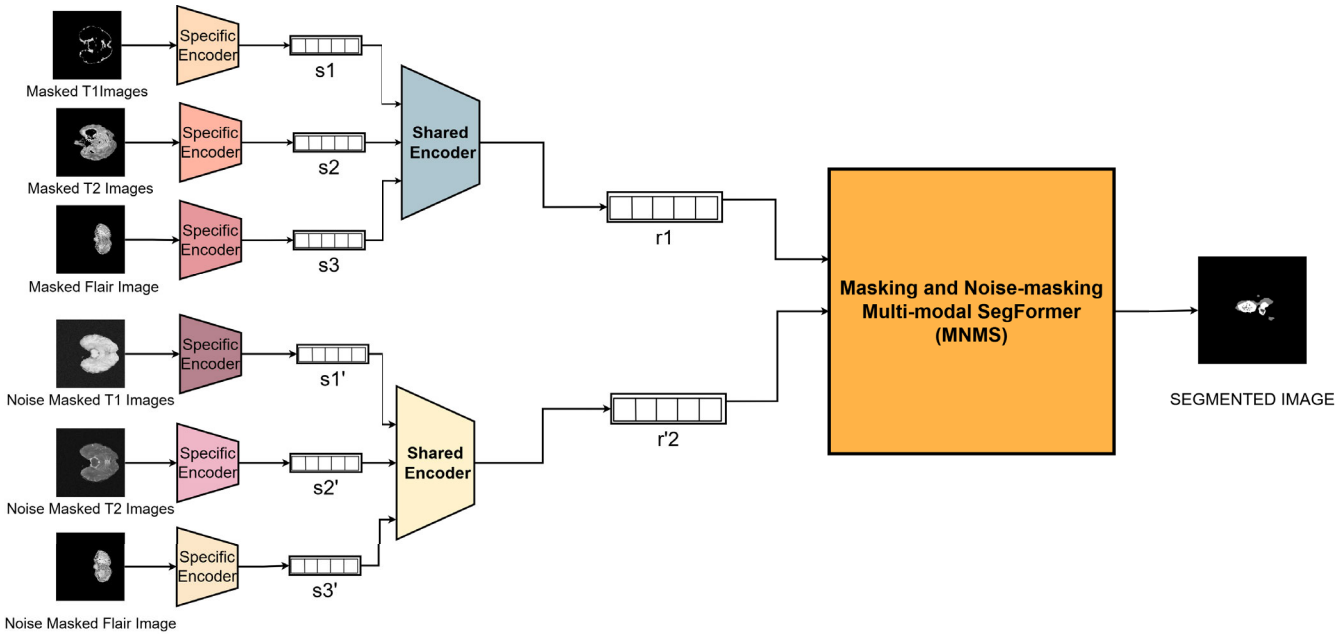

FIGURE 1. Overview of the proposed MNMS framework, which fuses window-masked and noise-masked T1, T2, and FLAIR MRI features using shared encoders and a SegFormer-based architecture to generate segmented brain tumor images.

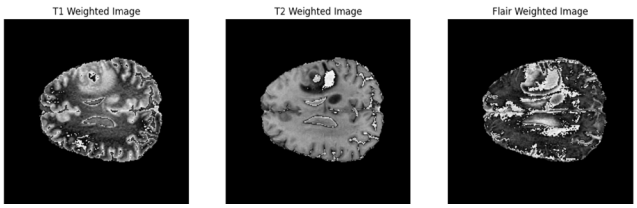

FIGURE 2. Sample slices of T1, T2 and FLAIR Modalities.

uniformity across modalities and align with the MNMS model input. C denotes the channel size which refers to the number of channels in the input data. A channel size of 3, the standard channel size for SegFormer-based models, is used to perform channel-wise normalization to ensure consistent scaling. Finally, the images are resized to a dimension of (256, 256, 3) and each modality is labelled in filenames depicting the modality type.

These comprehensive pre-processing steps ensure well-prepared input data for training the MNMS model, enabling effective handling of multimodal medical image segmentation and improving model performance on diverse MRI datasets involving T1, T2, and FLAIR modalities.

3) DATA CONVERSION AND AUGMENTATION

This work adopts a noise-masking module and a window-masking module, introduced during training. The noise masking module artificially introduces noise patterns that simulate real-world artifacts, helping the model learn to handle imperfect data effectively. Noise-masking is applied individually to T1, T2, and FLAIR modalities to account for the distinct characteristics of each. The window-masking module applies a random window mask which removes portions of the image, forcing the model to infer missing data.

In order to make feature extraction more resilient, the noise mask that is applied, forces the model to rely on surrounding information rather than being overly dependent on specific pixel values. For pixel-wise noise masking, a noise function $\eta(i, j)$ is defined for each pixel location (i, j), such that the masked image $I_{n-masked}$ for each input image I is:

$$I_{n-masked}(i, j) = \begin{cases} I(i, j) & \text{if } (i, j) \notin \Omega \\ \eta(i, j) & \text{if } (i, j) \in \Omega \end{cases} \tag{2}$$

where $I(i, j)$ is the original pixel intensity, Ω is the set of pixel locations selected for noise-masking, $\eta(i, j)$ represents noise applied to the selected pixels.

In T1-weighted images, noise masking helps simulate variations in anatomical structures; for T2-weighted images, it helps account for intensity variations due to fluid content; and in FLAIR images, noise-masking simulates artifacts often seen in clinical acquisitions.

So as to enhance the model's ability to generalize, Gaussian noise is applied to selected pixels, encouraging it to extract meaningful patterns even in the presence of missing data. For MNMS model, 15% noise level is used for noise-masking while training. These are small disturbances that slightly mask image details while maintaining most structural information. The noise function $\eta(i, j)$ is defined as:

$$\eta(i, j) \sim \mathcal{N}(\mu, \sigma^2) \tag{3}$$

where μ is the mean (typically zero) and σ^2 is the variance, controlling noise intensity. It simulates real-world sensor

and data acquisition noise, helping the model learn to ignore random fluctuations and focus on meaningful features, improving robustness in medical image segmentation.

Window-masking involves applying random window masks to remove portions of the image, forcing the model to infer missing data. Since a dataset of multiple MRI modalities is used, adaptive masking pushes the model to learn both modality-specific features and cross-modal shared features. This ensures the network doesn't overfit to just the unmasked data but instead focuses on deeper feature extraction. For pixel-wise window-masking, patches of pixels are removed, helping the model learn to infer missing parts from their surroundings, reinforcing global context understanding such that $I_{w-masked}$ for each input image I is:

$$I_{w-masked}(i,j) = \begin{cases} I(i,j) & \text{if } (i,j) \notin V \\ 0 & \text{if } (i,j) \in V \end{cases} \tag{4}$$

where $I(i,j)$ is the original pixel intensity and V is the set of pixel coordinates selected inside the window region for window-masking.

The window-masking strategy in our implementation uses fixed-size square patches of 20×20 pixels applied randomly to every MRI slice. These patches are masked at a random position in every slice during training so that the model is exposed to varying spatial arrangements of missing data. This fixed patch size was selected to balance between the loss of local detail and enough context for the model to pick up on important reconstruction.

By incorporating both Gaussian noise-masking and window-masking across T1, T2, and FLAIR modalities, the model is forced to rely on global context and cross-modal information to infer missing details - just like it would need to in real-world segmentation tasks. This strategy enhances its ability to handle missing or degraded inputs effectively, leading to more reliable feature extraction and improved segmentation performance.

Data augmentation techniques, such as rotation, flipping, cropping, and intensity shifts, are introduced to enhance generalization and prevent overfitting. These augmentations simulate variations in real-world MRI data, making the model more robust to unseen images.

C. MODALITY-SPECIFIC ENCODER

After applying noise-masking and window-masking, the next critical step involves extracting the features from the masked images. A modality-specific encoder is allocated for each modality (T1, T2 and FLAIR) of each input data type (window-masked images and noise-masked images). Each image undergoes the same set of processes designed for the encoder. Let I_m represent the input MRI images, where I_{T1}, I_{T2} and I_{FLAIR} correspond to the T1-weighted, T2-weighted, and FLAIR modalities, respectively. These images undergo modality-specific encoding, denoted as $f_m(I_m)$, where the function $f_m(\cdot)$ extracts modality-dependent feature representations.

The encoder takes a 2-channel input formed by concatenating the raw modality image and its binary mask. It comprises three hierarchical blocks, each with two 3×3 convolutional layers (padding $= 1$). The first block operates at full resolution with 64 filters, while the second and third blocks downsample the input (stride $= 2$) and increase the channels to 128 and 256, respectively. Each convolution is followed by instance normalization and Leaky ReLU activation (slope $= 0.2$), with Dropout2D ($p = 0.2$) applied after each block for regularization. An attention module is applied to the final feature map, consisting of two 1×1 convolutions: the first reduces channels from 256 to 64 followed by Leaky ReLU, and the second restores it to 256 using a sigmoid to generate an attention map. This map re-weights the final features via element-wise multiplication. A final 1×1 convolution projects the attention-enhanced features to an output of shape [batch size, 256, H/4, W/4].

This step ensures that the model captures unique tissue contrasts, anatomical structures, and pathological markers inherent to each MRI modality.

1) FEATURE EXTRACTION

The modality-specific encoder extracts features from the masked inputs I_m, producing feature maps F_m, where:

$$F_m = f_m(I_m) \tag{5}$$

Each feature map F_m contains information relevant to its corresponding modality:

- F_{T1} highlights soft tissue contrasts in T1-weighted images.
- F_{T2} emphasizes fluid-sensitive characteristics in T2-weighted images.
- F_{FLAIR} captures lesion-specific patterns useful for identifying edema.

By compelling the encoder to learn robust, context-aware representations, the noise-masking mechanism ensures that the extracted features remain resilient to noise and artifacts, enhancing generalization.

2) FEATURE REPRESENTATION AND ALIGNMENT

Once the features F_m are extracted from I_m the next step is to align and represent them in a common space. The encoder applies a transformation function $g(F_m)$ to normalize the feature distributions across modalities:

$$F'_m = g(F_m) \tag{6}$$

where F'_m represents the aligned feature maps from different MRI modalities. The final set of aligned features $F_{aligned}$ is obtained as:

$$F_{aligned} = [F'_{T1}, F'_{T2}, F'_{FLAIR}] \tag{7}$$

This step ensures that modality-specific features are effectively utilized in the segmentation process. By jointly processing these aligned features, the model gains a deeper understanding of multimodal data, improving segmentation accuracy and providing better segmentation results.

D. SHARED ENCODER NETWORK

After the features are obtained from the modality-specific encoders, the next step is to combine and process them within a single framework. To accomplish this, a shared encoder network is introduced, allowing the model to effectively merge features from T1, T2, and FLAIR images, including window-masked and noise-masked versions, into a single feature space.

The shared encoder is a convolutional neural network designed for learning shared representations from multi-modal 2D medical images, such as T1, T2, and FLAIR MRI slices. Each modality captures different anatomical or pathological characteristics, and fusing them effectively is essential for comprehensive understanding. This encoder leverages early fusion by accepting stacked input channels from different modalities, enabling joint feature learning across modalities. It employs a multi-scale convolutional pipeline to extract spatial features at varying resolutions, allowing the network to capture both fine-grained and coarse structural patterns. An attention mechanism is integrated at the bottleneck to adaptively weigh the most informative feature regions, thereby enhancing interpretability and discriminative power. The reconstruction pathway uses transposed convolutions to upsample and restore spatial dimensions, preserving crucial modality-aware information. This design is well-suited for tasks such as tumor or lesion segmentation, where capturing cross-modal correlations improves the model's ability to localize and differentiate pathological regions effectively.

This approach makes sure that the model benefits from cross-modal information while preserving important modality- specific details, which are crucial for accurate segmentation.

1) FEATURE FUSION FOR SHARED ENCODING

In multimodal medical image segmentation, to improve segmentation accuracy combining features from different MRI modalities is a vital step. Each modality provides distinctive tissue contrasts, and merging them improves the model's ability to identify tumors and anatomical structures more effectively.

To attain this, a shared encoder network, denoted as E_s, processes obtained features from T1, T2, and FLAIR images, creating a robust multimodal representation:

$$F_{shared} = E_s([F_{T1}, F_{T2}, F_{FLAIR}]) \tag{8}$$

where F_{shared} represents the single feature space, ensuring that information from all modalities is effectively combined for better segmentation performance.

2) SHARED ENCODER PROCESSING

The shared encoder extracts two types of feature streams -Window-masked Features ($F_m^{\text{w-masked}}$) and Noise-Masked Features ($F_m^{\text{n-masked}}$).

Both streams are independently passed through their respective shared encoders, denoted as $E_s^{\text{w-masked}}$ and

Algorithm 1 Masked and Noise-Masked Multimodal Medical Image Segmentation

Input: BraTS2020 dataset containing T1, T2, and FLAIR MRI modalities.
Output: Segmented brain MRI image, S.

1: Convert.nii images to.png format with shape (256, 256, 3).
2: Create dataloaders for each modality.
3: Generate two sets of images for each modality:
- **Gaussian noise:**

$$I_{n-masked}(i,j) = \begin{cases} I(i,j), & \text{if } (i,j) \notin \Omega \\ \eta(i,j), & \text{if } (i,j) \in \Omega \end{cases}$$

- **Window masking:**

$$I_{w-masked}(i,j) = \begin{cases} I(i,j), & \text{if } (i,j) \notin V \\ 0, & \text{if } (i,j) \in V \end{cases}$$

4: **for** each modality-specific encoder **do**
5: Extract features separately for noise-masked and window-masked images:

$$F_{T1} = f_m(I_{T1}), \quad F_{T2} = f_m(I_{T2})$$
$$F_{FLAIR} = f_m(I_{FLAIR})$$

6: **end for**
7: Fuse features from all modalities using shared encoders:

$$F_{shared,noise} = E_s(F_{T1,noise} \oplus F_{T2,noise} \oplus F_{FLAIR,noise})$$
$$F_{shared,mask} = E_s(F_{T1,mask} \oplus F_{T2,mask} \oplus F_{FLAIR,mask})$$

8: Fuse shared features: $F_{aligned} = F_{shared,noise} \oplus F_{shared,mask}$
9: **for** each set of aligned features $F_{aligned}$ **do**
10: $Patches \leftarrow P_{embed} = \text{Conv}(F_{aligned}; k, s, p)$
11: $Feature\ Tokens \leftarrow$ Apply linear projections to patches
12: **end for**
13: **while** $t < \max_{\text{iter}}$ **do**
14: Process patches using hierarchical encoder
15: Compute multi-head self-attention (MHSA):

$$\text{MHSA}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

16: Fuse multi-scale features using MLP-based feature aggregation
17: **end while**
18: **for** each feature map **do**
19: $Upsampled\ features \leftarrow Upsample(MLP(F_1 \oplus F_2 \oplus \cdots \oplus F_n))$
20: Compute segmentation map:

$$S(i,j) = \frac{e^{z_c}}{\sum_{k=1}^{C} e^{z_k}}$$

21: Append S to output
22: **end for**
23: **return** segmented MRI image S.

$E_s^{\text{n-masked}}$, for further processing:

$$F_s^{\text{w-masked}} = E_s^{\text{w-masked}}(F_m^{\text{w-masked}}) \tag{9}$$

$$F_s^{\text{n-masked}} = E_s^{\text{n-masked}}(F_m^{\text{n-masked}}) \tag{10}$$

This dual-encoder structure enables the model to learn patterns from both window-masked and noise-masked inputs, avoiding dependency on unstable, pixel-level details. Instead, the model forms stronger, higher-level feature maps which are crucial for segmenting complex medical images. Also, the model becomes more cautious and better calibrated, producing more reliable segmentation maps with better confidence scores.

3) CROSS-MODALITY FEATURE ALIGNMENT

After processing shared features, the next step is cross-modality feature alignment, ensuring that the learned features from $F_s^{\text{w-masked}}$ and $F_s^{\text{n-masked}}$ are mapped into a common representational space. This alignment is achieved using a transformation function $h(\cdot)$ that reduces the modality gap:

$$F_{CMFA} = h([F_s^{\text{w-masked}}, F_s^{\text{n-masked}}]) \tag{11}$$

where F_{CMFA} represents the final multimodal feature representation. By combining features from both window-masked and noise-masked streams, the model improves its generalization ability across varied datasets, leading to higher segmentation accuracy.

Algorithm 1 outlines the MNMS framework. The input consists of T1, T2, and FLAIR MRI modalities from the BraTS2020 dataset, and the output is the segmented brain MRI image S. Initially,.nii scans are converted to.png format with a size of (256 × 256 × 3) and augmented using Gaussian noise and window-masking. Gaussian noise masking adds noise to pixels outside a specified region, while window masking zeroes out selected patches in the image. Each masked image is passed through its corresponding modality-specific encoder f_m to extract feature vectors $F_{T1}, F_{T2}, F_{FLAIR}$. These feature vectors are then passed into shared encoders to perform cross-modal feature fusion.

The shared encoder outputs two fused feature representations - one for noise-masked inputs and the other for window-masked inputs. These are then combined to obtain a final aligned feature set $F_{aligned}$, as indicated in line 8. These features are patch-embedded, tokenized, and processed using a hierarchical transformer with Multi-Head Self-Attention (MHSA), as described in Eq. (15), enabling the model to capture long-range dependencies and contextual information. Finally, multi-scale features are aggregated via MLPs, upsampled, and used to generate the final segmentation map $S(i, j)$ through a softmax-based classification, as described in line 20. The segmented image S is then appended to the output.

E. ARCHITECTURE OF MNMS

After cross-modality feature alignment, where features from window-masked and noise-masked streams are mapped into a common representational space, the next step is to fine-tune the MNMS model for precise brain MRI segmentation. The aligned features, $F_{aligned}$, now serve as the input to the MNMS architecture, allowing it to leverage both window-masked and noise-masked information for robust segmentation. MNMS, a transformer-based architecture, is well-suited for this task due to its ability to efficiently capture local and global contextual dependencies, ensuring improved segmentation accuracy even in challenging real-world clinical scenarios. MNMS acts as the backbone of the entire pipeline, offering multi-scale feature extraction, lightweight decoding, and robust performance across different image modalities.

1) PATCH EMBEDDING

In order to process the input feature maps efficiently, MNMS employs an overlapping patch embedding mechanism, which converts the aligned features into smaller patches of size $P \times P$ for further processing. The formula for patch embedding can be written as:

$$P_{embed} = \text{Conv}(F_{aligned}; k, s, p) \tag{12}$$

where $F_{aligned}$ represents the final multimodal representation, k is the kernel size of the convolution, which defines the patch size, s is the stride, determining how far the kernel moves during the convolution, which is usually set to match the patch size, and p is the padding, sometimes used to maintain spatial dimensions.

Unlike traditional ViTs that use non-overlapping patches, MNMS ensures spatial continuity by utilizing overlapping regions, preserving important structural details in brain MRI images. Each patch is projected into a lower-dimensional space using convolutional layers, reducing computational complexity while retaining critical spatial information. This patch embedding step prepares the input features for hierarchical feature extraction in the encoder.

2) NON-POSITIONAL ENCODING

In traditional transformer models like ViTs, positional encoding is used to inject spatial information into the input patches. This is because transformers process input data as sequences of patches and lack the inherent ability of CNNs to understand spatial relationships. Unlike transformers, which rely on positional encoding to capture spatial information, MNMS removes positional encoding entirely. This design choice enhances the model's robustness to variations in image scale and structure, making it more suitable for handling diverse medical imaging modalities without relying on fixed spatial layouts.

3) HIERARCHICAL ENCODER

After the patch embedding stage, the encoded patch embeddings are processed by the Mix Transformer (MiT) encoder of the MNMS, which extracts multi-scale hierarchical features while maintaining fine-grained local information and long-range dependencies. The MiT encoder gradually reduces

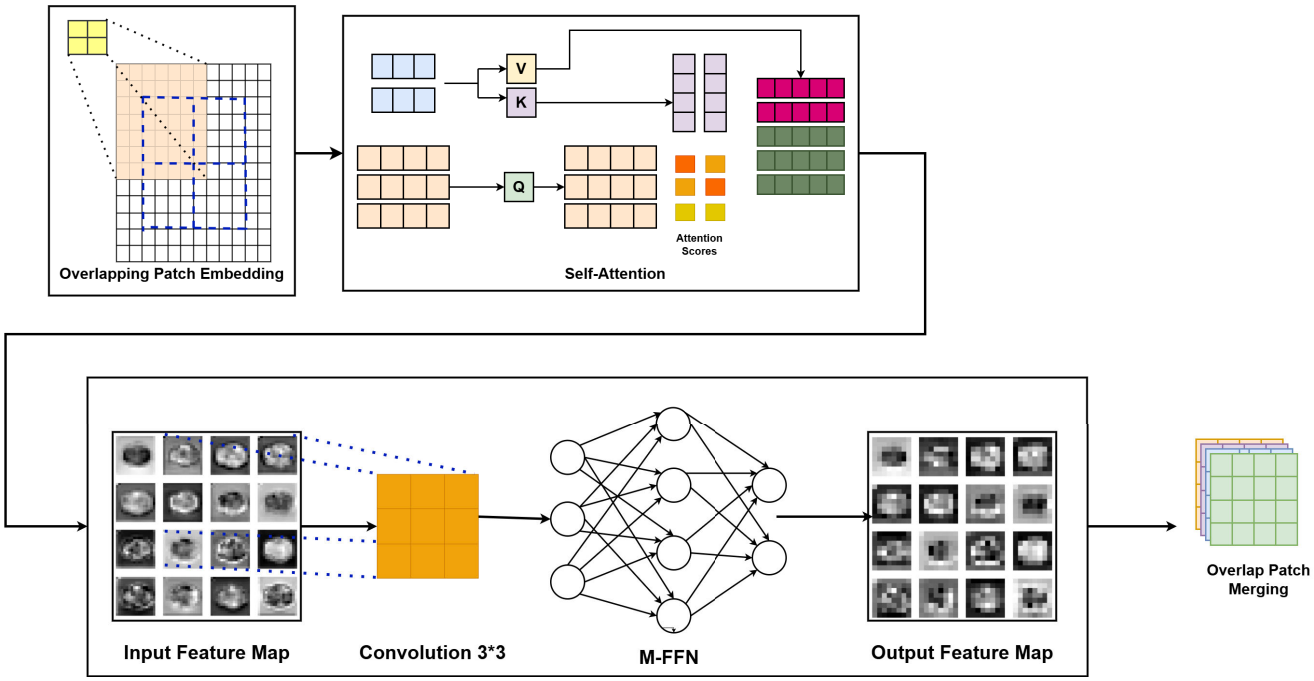

FIGURE 3. Overview of the MNMS architecture, illustrating overlapping patch embedding, self-attention computation, and multilayer feed-forward network (M-FFN) operations with convolution, followed by overlap patch merging.

spatial resolution across several stages, akin to a CNN-based feature pyramid, in contrast to conventional CNNs, which depend on fixed receptive fields.

Each level of input data management in the hierarchical structure of the MiT encoder is designed to collect features at different scales. The 4×4 patch size of the first layer allows the model to collect more geographical relationships and contextual information. In the following layers, the patch size is decreased to 2×2, which enables the encoder to focus on smaller details and local structures. This multi-scale approach ensures that high-level semantic information and low-level spatial data are acquired efficiently, leading to precise tumor segmentation in brain MRI scans.

To enhance feature representation across a variety of MRI modalities, the encoder additionally uses a cross-modality attention method. This technique allows information from one modality to improve and enhance another during segmentation by dynamically weighting important characteristics from multiple MRI sequences. This integration increases the overall segmentation accuracy of the model by exploiting cross-modal information. The attention mechanism can be defined mathematically as follows:

$$Q = W_q \cdot F_{T1}, K = W_k \cdot F_{T2}, V = W_v \cdot F_{FLAIR} \tag{13}$$

where W_q, W_k, W_v are learnable weight matrices applied to various MRI modalities. This makes it possible for the MNMS architecture to concentrate on certain anatomical features, improving segmentation performance by utilizing information on complementary modalities.

In order to efficiently process the fused multimodal features, MNMS employs transformer layers, consisting of MHSA and MLP blocks. MHSA enables the model to capture long-range dependencies by enabling different regions of the image to interact, making it adept at understanding global spatial relationships and localized tumor structures. The self-attention operation is formulated as:

$$\text{MHSA}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{14}$$

where d_k is the scaling factor for stabilizing gradient updates. Here, Q (query), K (key), and V (value) represent feature embeddings that are processed through self-attention. The model can highlight important tumor locations in MRI scans according to the softmax function, which gives higher weights to the most relevant features.

Following self-attention, a 3×3 convolutional layer is applied to the returned feature maps, enhancing local spatial linkages while preserving the global context acquired during self-attention. The fine-grained characteristics of the tumor border are preserved at this stage.

After refinement, the features are processed by a Mixed Feed-Forward Network (M-FFN), which adds non-linearity and transforms them to enhance trained representations for more precise segmentation. The GeLU activation function that separates the two linear layers that comprise the M-FFN looks like this:

$$\text{M-FFN}(X) = \text{L}_2\left(\sigma\left(\text{L}_1(X)\right)\right) \tag{15}$$

where σ is the GeLU activation function, and L_1 and L_2 are the two linear layers. The model can represent intricate

feature interactions while retaining computational efficiency due to this framework.

Finally, the produced feature maps are subjected to Overlap Patch Merging, a method that reduces spatial resolution while preserving significant border features. Ensuring that the hierarchical structure of the encoder maintains its effectiveness while maintaining spatial continuity results in better-segmented images.

By combining these techniques, MNMS effectively learns both modality-specific and shared representations, ensuring robustness and enhancing segmentation accuracy across a range of MRI datasets.

4) LIGHTWEIGHT DECODER

Following hierarchical feature extraction, MNMS uses a lightweight MLP-based decoder to generate the final segmentation mask.

MNMS generates the final segmentation mask using a lightweight MLP-based decoder, which effectively combines multi-scale feature maps utilising a straightforward yet powerful MLP structure, in contrast to conventional transformer-based decoders that depend on complex upsampling procedures. This method preserves good segmentation precision while drastically lowering computational overhead. To ensure that even the smallest cancer patches are precisely captured, the decoder reconstructs a high-resolution segmentation output after analysing the multi-resolution feature maps that the encoder extracted.

For a lightweight MLP-based decoder, the decoding process can be mathematically represented by a simple yet effective formula:

$$S = Upsample\left(MLP\left(F_1 \oplus F_2 \oplus \cdots \oplus F_n\right)\right) \tag{16}$$

where S is the final high-resolution segmentation map, $F_1, F_2, \ldots, F_n$ are the multi-scale feature maps extracted from the encoder, $\oplus$ is the concatenation operation along the channel dimension, *MLP* is a multilayer perceptron that refines the fused features and *Upsample* is an operation to restore the spatial resolution.

5) SEGMENTATION OF BRAIN TUMOR FROM MEDICAL IMAGES

The final segmentation map $S(i, j)$ is obtained by applying a softmax activation function to the output of the last decoder layer. This ensures that the network produces a probability distribution over C possible classes for each pixel location (i, j). The softmax function is defined as:

$$S(i, j) = \frac{e^{z_c}}{\sum_{k=1}^{C} e^{z_k}} \tag{17}$$

where z_c is the logit (raw score) computed by the model for class c at pixel (i, j) and S_c is the total number of segmentation classes. The denominator ensures that the sum of probabilities for all S_c classes at a given pixel location equals 1, making the output interpretable as a probability distribution.

This softmax output is then used to generate the final segmentation mask, where each pixel is assigned the class with the highest probability:

$$\hat{c}(i, j) = \arg\max_{c} S(i, j) \tag{18}$$

where $\hat{c}(i, j)$ represents the predicted class at pixel (i, j). This method improves model robustness in multimodal medical imaging scenarios, handles class boundaries efficiently, and guarantees accurate and seamless segmentation.

F. EVALUATION AND OPTIMIZATION TECHNIQUES

Accuracy, overlap, and the model's ability to identify cancer regions are essential factors for evaluating segmentation performance. Further refining the final segmentation results through post-processing and fine-tuning procedures guarantees greater accuracy and consistency.

1) DICE LOSS

The dice loss function is a widely used evaluation metric to rate the overlap between the segmentation maps of the ground truth mask and the predicted mask. Dice loss is helpful in detecting small objects and class imbalances as it emphasizes the accuracy of smaller regions like tumors.

A combination of Dice Loss and Cross-Entropy Loss is often used to discuss class imbalance and improve segmentation accuracy of the model.

2) DSC (DICE COEFFICIENT)

The Dice Coefficient (DSC) is a commonly used evaluation metric for evaluating segmentation accuracy. DSC calculates the similarity between the ground truth mask and predicted segmentation mask and provides a statistical analysis of the performance of the model with respect to identifying the regions of interest with at most accuracy.

A DSC value closer to 0 indicates poor overlap between the actual and predicted tumor regions, while a DSC value closer to 1 suggests better segmentation accuracy.

3) IoU (INTERSECTION OVER UNION)

The Intersection over Union (IoU), also known as the Jaccard Index, calculates the ratio of the intersection area to the union of segmentation and ground truth mask regions in order to decide how well the segmented output matches the ground truth.

Higher the IoU value, better is the segmentation performance, with an ideal value of 1 denoting perfect segmentation.

4) PRECISION AND RECALL

In order to assess the model's capacity to correctly define tumor pixels while reducing false positives and false negatives, precision and recall are key variables.

The percentage of accurately predicted cancer pixels out of all expected tumor pixels is known as precision (positive predictive value). In precison, True Positives (TP)

represents tumor pixels correctly identified as tumors, and False Positives (FP) represents tumor pixels mistakenly classified as tumors.

Recall (Sensitivity or True Positive Rate) shows the percentage of real tumor pixels that the model accurately identified. In recall, False Negatives (FN) represents tumor pixels that are wrongly categorised as non-tumors and TP represents tumor pixels that are correctly recognised as tumors.

A high precision indicates fewer false positives, while a high recall indicates fewer false negatives. To balance both metrics, the F1-score is often used. This provides a single score that considers both precision and recall, making it a useful measure of overall segmentation performance.

5) VISUAL INSPECTION

Quantitative metrics such as DSC, IoU, precision and recall are known to provide valuable information about the model's performance. While these metrics play a crucial role in evaluating the model's performance, visual inspection is a much-needed step to evaluate segmentation quality. Experts can assess the accuracy of the prediction by overlapping the predicted masks on the original images and by comparing the tumor boundaries captured. Blurry edges, missing tumor regions or partial volume effects may be overlooked by automatic evaluation metrics. These issues can be solved by giving special attention to certain important areas, mainly tumor regions, to make sure the segmentation boundary lines align accurately with the original anatomical structures. Experts make sure that the model has predicted the tumor and non-tumor regions accurately in order to ensure the results are precise and meaningful. Visual inspection also helps to determine whether further fine-tuning is required to improve the segmentation accuracy of the model.

6) FINE-TUNING

After visual inspection, the fine-tuning procedure takes place in order to optimize the MNMS model with the help of aligned and fused feature representations. The model is usually trained using a loss function which can either be dice loss or a combination of dice loss and cross-entropy loss. Fine-tuning tunes MNMS's parameters so that the model can learn segmentation patterns from the specified MRI dataset. As a result, the differentiation between tumor and non-tumor regions is improved and much more precise and reliable segmentation outcomes are received. When the fine-tuned MNMS models is tested on the testing set, quantitative assessment is taken. Integration of MNMS into the architecture pipeline ensures the model fully explores the multimodal feature space to guarantee robust segmentation, even in the presence of noise and other external factors. Fine-tuning ensures there is significant improvement in the model's generalization ability on diverse medical datasets.

7) POST-PROCESSING

Finally, post-processing takes place to ensure that the model's predictions are refined, free from external disturbances and clinically significant. This final phase is crucial in enhancing the accuracy and overall reliability of the segmentation results produced by the model. This phase addresses common challenges such as artifacts, fragmented regions, and uncertain boundaries, aiding to create more precise and reliable segmentation masks. Post-processing step ensures that the segmented regions are rightly aligned with the actual anatomical structures. This ensures that the output produced is reliable and meaningful, increasing its chances of being suitable for real-world clinical applications.

IV. RESULTS

This section presents the experimental results of the proposed approach for multimodal brain tumor image segmentation using MNMS. A detailed analysis of the experimental outcomes offers a clear understanding of the model's effectiveness and its performance across various segmentation metrics.

The experiments were conducted on a system equipped with an Apple M1 chip, 8 GB of RAM, and a 64-bit macOS operating system. Additionally, a NVIDIA A100 Tensor Core GPU with 40 GB of memory provided the computational power required for intensive training and data processing tasks. The model was implemented using Keras, TensorFlow, PyTorch, and OpenCV libraries, leveraging these frameworks to streamline data handling, model training, and image analysis.

Figure 4 highlights a comparative visualization of brain MRI slices across three modalities - T1-weighted, T2-weighted, and FLAIR, alongside their corresponding ground truth masks and the segmented outputs generated by the MNMS model. The figure clearly illustrates how the MNMS framework effectively identifies and segments tumor regions by drawing on complementary features from multiple modalities. Despite differences in intensity and texture between T1, T2, and FLAIR images, the segmented outputs closely align with the ground truth, capturing small and large tumor regions with high precision. This demonstrates the model's ability to integrate cross-modality information and effectively learn global spatial relationships and fine-grained tumor boundaries, contributing to robust and accurate brain tumor image segmentation.

Figure 5 presents a comparative analysis of segmentation performance across different models - Swin-UNet, SegFormer, and the proposed MNMS framework against the ground truth for brain MRI slices. The results clearly highlight the superior performance of MNMS in accurately identifying tumor regions. Swin-UNet struggles with under-segmentation, often missing crucial tumor boundaries, while SegFormer shows relatively better segmentation but tends to produce fragmented regions, failing to capture the full tumor structure. Compared to baseline models, MNMS

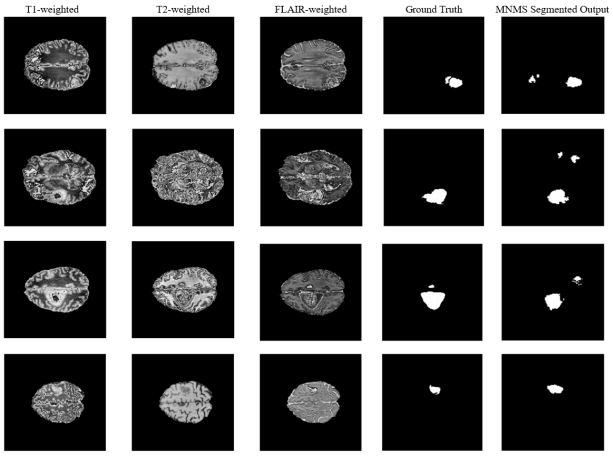

FIGURE 4. Comparison between image slices from different modalities and the segmented output using MNMS.

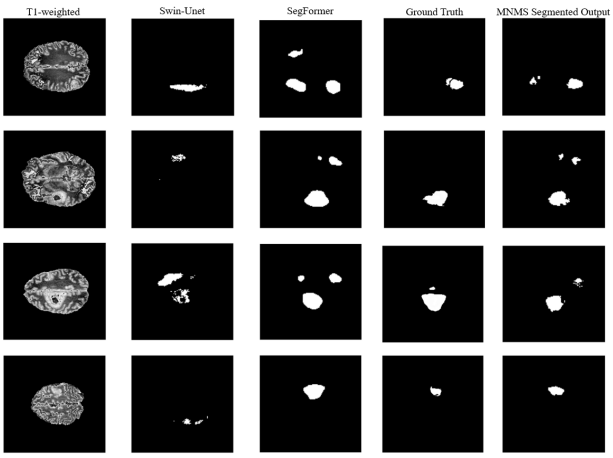

FIGURE 5. Comparison of the segmentation performance of different models.

demonstrates superior accuracy in delineating tumor regions. This can be attributed to its cross-modality attention mechanism and hierarchical feature extraction, which allow the model to capture both global contextual information and fine-grained anatomical details. These capabilities make MNMS particularly effective in handling the complexities of multimodal medical image segmentation.

To further evaluate the model's performance at different training stages, Figure 6 presents the confusion matrices at 50 epochs, 75 epochs, and 100 epochs. These matrices offer a detailed breakdown of the model's classification results, showing the distribution of TPs, FPs, TNs, and FNs for each class. The values correspond to the number of pixels classified across the entire test set, reflecting the model's performance at the voxel level rather than per image. The progressive improvement in the confusion matrices highlights how the MNMS model becomes more accurate and reliable as training advances.

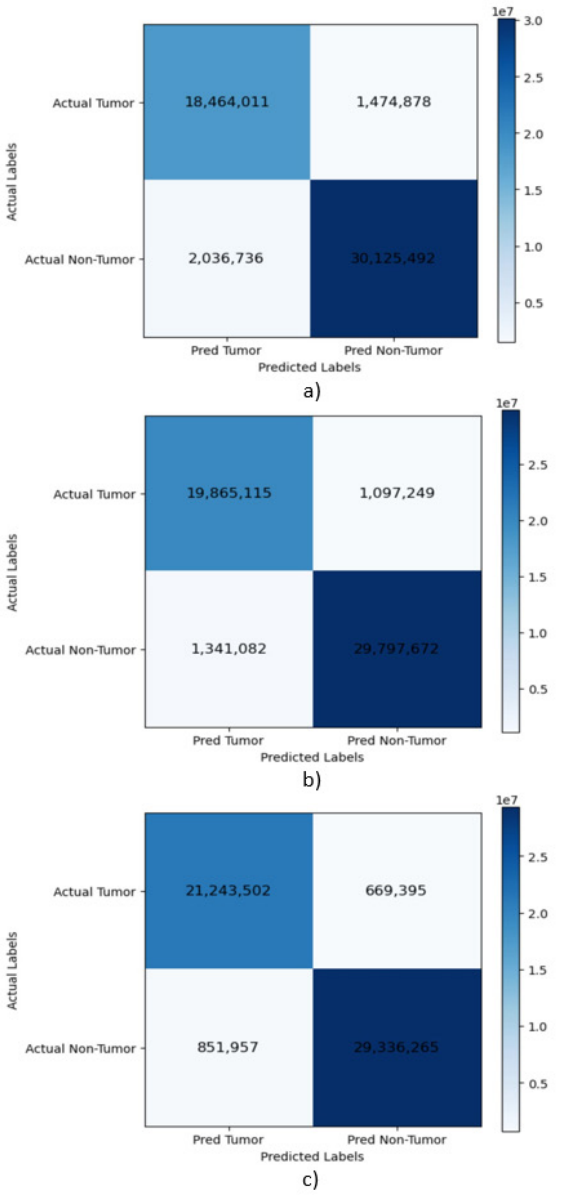

FIGURE 6. The confusion matrices for the segmentation results of the proposed model for a) 50 epochs b) 75 epochs c) 100 epochs.

At 50 epochs, the confusion matrix indicates that the model is still in the early stages of learning. Misclassifications are more frequent, suggesting that the feature representations are not yet well-optimized. This is reflected in relatively lower recall and precision values across multiple classes.

By 75 epochs, there is a noticeable improvement in classification accuracy. The number of false positives and false negatives decreases, suggesting that the model has learned better feature representations. The diagonal dominance in the matrix is more pronounced, indicating stronger agreement between predicted and actual labels.

At 100 epochs, the model achieves its best performance, as evidenced by an even higher concentration of

correct classifications along the diagonal. The number of misclassified samples is minimized, and both sensitivity and specificity show improvement. However, if the off-diagonal values do not reduce significantly between 100 and 150 epochs, it may indicate a saturation point, where additional training does not yield substantial performance gains.

The progression of these confusion matrices highlights the importance of training duration in optimizing model performance. It also provides insight into whether additional training epochs contribute to meaningful improvements or if early stopping should be considered to prevent overfitting.

The accuracy plots shown in Figure 7 illustrate the progression of training and validation accuracy over 50, 75, and 100 epochs, respectively. Across all epochs, both training and validation accuracy exhibit a steady increase, with no signs of significant instability or abrupt fluctuations, indicating stable learning and effective gradient updates. The validation accuracy closely follows the training accuracy in all three cases. While there is a slight gap between the two curves, it narrows as training progresses, suggesting that the model generalizes well without overfitting.

As shown in Figure 7a, at 50 epochs, the model reaches approximately 93% training accuracy and 89% validation accuracy and as shown in Figure 7b by 75 epochs, the training accuracy and validation accuracy continue to rise, nearing 96% and 92%, respectively. Finally, at 100 epochs, as shown in Figure 7c, the model stabilizes around 98% training accuracy and 94% validation accuracy.

The training and validation loss curves, shown in Figure 8, illustrate the model's learning dynamics over 50, 75, and 100 epochs, respectively. Across all epochs, both the training and validation loss exhibit a consistent downward trend, confirming that the model is effectively learning to minimize error during training. As shown in Figure 8a, at 50 epochs, the loss decreases sharply in the initial epochs before gradually stabilizing, ending at approximately 0.5 for training loss and 0.9 for validation loss and by 75 epochs, shown in Figure 8b, the losses further decline, with training and validation losses converging closer, around 0.4 and 0.6, respectively. At 100 epochs, shown in Figure 8c, the model achieves its lowest recorded losses: $\sim$ 0.3 for training loss and $\sim$ 0.5 for validation loss.

The training results presented in Table 4 demonstrate the progressive improvement of the proposed MNMS model over 100 epochs. Notably, the model's DSC increased from 0.8922 at epoch 50 to 0.9341 at epoch 100, indicating enhanced segmentation accuracy. Concurrently, the IoU rose from 0.8402 to 0.9209. The model's precision and recall also showed steady growth, achieving 0.9588 and 0.9538 by epoch 100, respectively. These improvements directly influenced the F1-score, which climbed from 0.9125 to 0.9567. Additionally, the training and validation loss consistently decreased, stabilizing at 0.23% and 0.41% at epoch 100, further signifying effective learning and reduced overfitting. These results highlight the efficacy of the proposed MNMS approach in delivering high segmentation accuracy with minimal loss, making it a robust solution for multimodal medical image segmentation tasks.

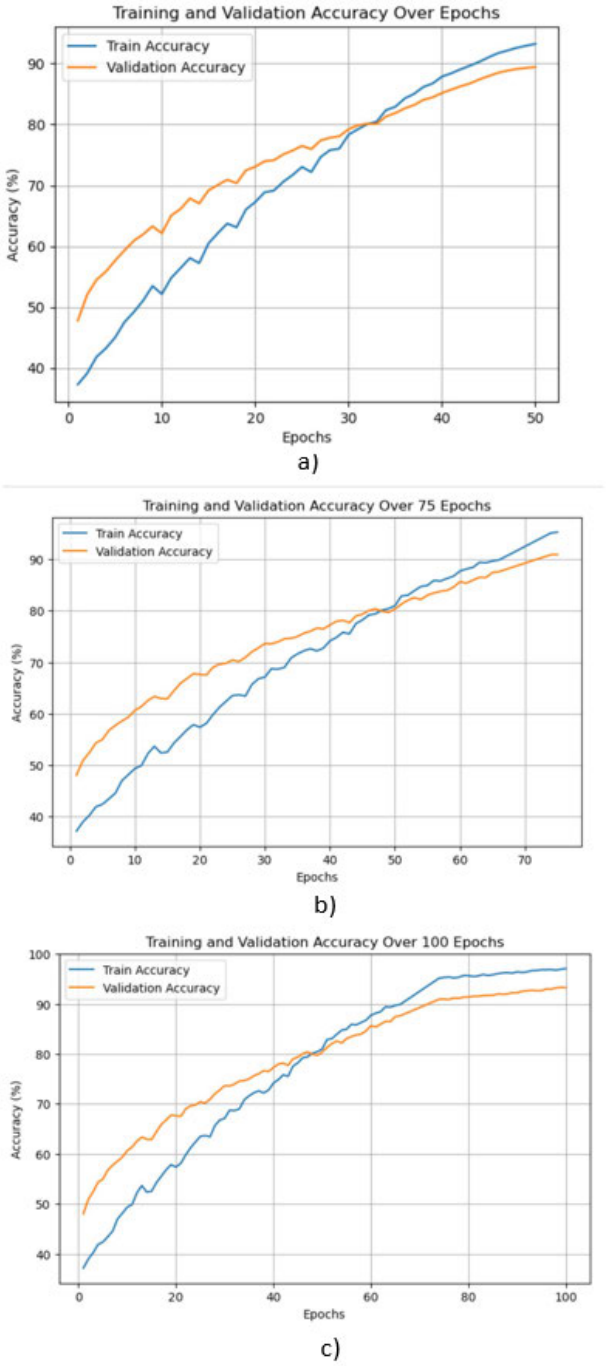

FIGURE 7. The training and validation accuracy scores of the proposed model over a) 50 epochs b) 75 epochs c) 100 epochs.

Table 5 presents the DSC comparison between the proposed MNMS model and state-of-the-art segmentation models, including 3DU-Net, DEU-Net, UNETR, and 3D Attention U-Net. The MNMS model achieves the highest DSC score of 0.9341, significantly outperforming the

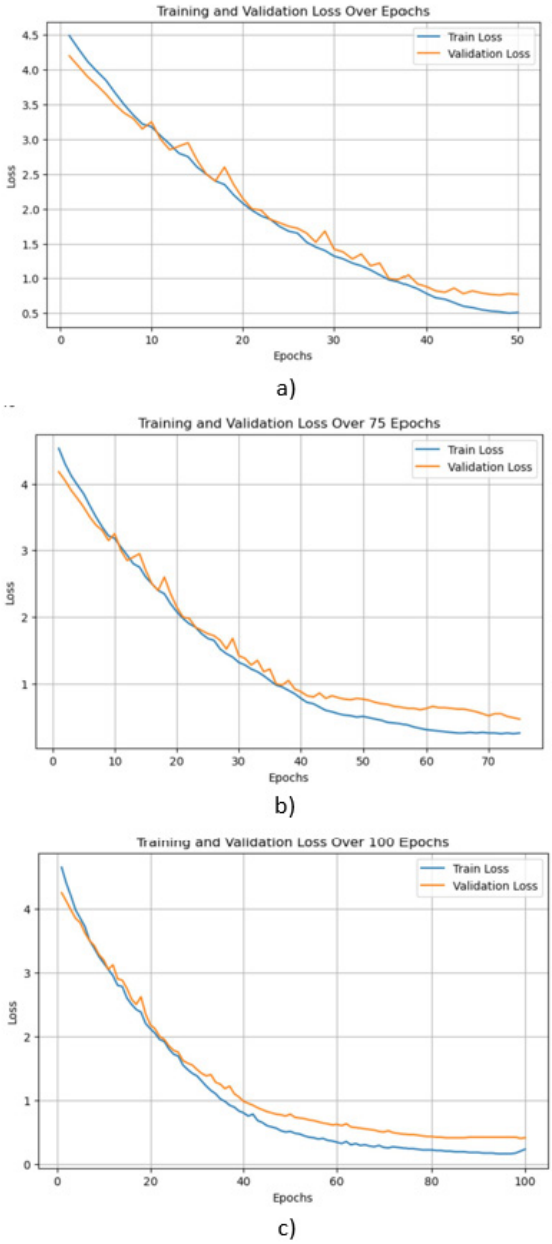

FIGURE 8. The training and validation loss scores of the proposed model over a) 50 epochs b) 75 epochs c) 100 epochs.

other methods, with DEU-Net being the closest competitor at 0.8422. This significant improvement underscores the effectiveness of the MNMS framework in accurately segmenting brain tumor regions. By leveraging multimodal feature fusion, noise-masking strategies, and cross-modality attention mechanisms, MNMS proves capable of capturing fine-grained spatial details while seamlessly integrating complementary information from multiple MRI modalities. These results establish MNMS as a new benchmark in multimodal medical image segmentation, highlighting its robustness and precision.

TABLE 4. Training results of the proposed model.

Metric	Epoch 50	Epoch 75	Epoch 100
Accuracy (%)	93.26	95.32	97.08
Training Loss (%)	0.51	0.26	0.23
Validation Loss (%)	0.77	0.57	0.41
DSC	0.8922	0.9262	0.9341
IoU	0.8402	0.8907	0.9209
Precision	0.9007	0.9368	0.9588
Recall	0.9260	0.9477	0.9538
F1-score	0.9125	0.9412	0.9567

TABLE 5. DSC comparison between state-of-the-art methods and the proposed model.

Method	DSC
3DU-Net [18]	0.8179
DEU-Net [21]	0.8422
UNETR [31]	0.8208
3D Attention U-Net [32]	0.8346
Proposed MNMS	**0.9341**

Despite the strong performance of the MNMS framework, it is important to recognize certain limitations that may affect its large-scale applicability. The model's evaluation was conducted completely on the BraTS 2020 dataset, which, while being comprehensive and widely used, may not fully reflect the complexity present in real-world clinical scenarios. The integration of noise-masking and window-masking strategies enhance robustness, but also adds computational overhead, which may limit its deployment in certain scenarios where there is a resource constraint. A critical constraint is the model's dependency on the availability of all three MRI modalities (T1, T2, and FLAIR) during training to fully capture complementary and cross-modal information. However, in real-world clinical settings, such complete multimodal datasets may be difficult to obtain due to scanner limitations, acquisition failures, or cost constraints. Although the ablation study (Section V) indicates that the MNMS model can still work when trained on a single modality, the accuracy decreases noticeably - from 97.08% with complete modalities to 77.19% with FLAIR-only training - highlighting the significance of multimodal integration. This limitation suggests a need for future enhancements such as modality dropout during training, modality hallucination techniques to synthetically generate missing modalities, or domain adaptation to increase generalizability. Additionally, the use of fixed noise levels and window-masking ratios may reduce adaptability across datasets with varying imaging characteristics. Even though the MNMS framework achieves high segmentation accuracy across standard quantitative metrics, validation on real-time clinical data and feedback from radiologists remain essential to fully assess its diagnostic utility and practical relevance.

V. ABLATION STUDY

To evaluate the contribution of individual MRI modalities in our segmentation pipeline, an ablation study was conducted

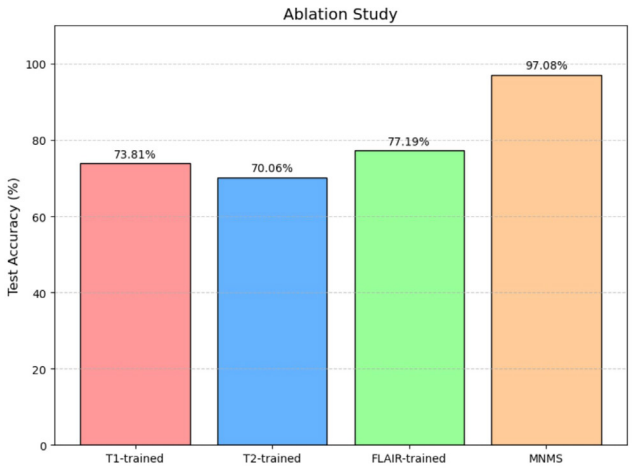

FIGURE 9. Ablation study comparing the test accuracy of the proposed MNMS model trained on individual MRI modalities (T1, T2, FLAIR) versus the full multimodal input.

wherein the proposed MNMS model was trained separately on T1-only, T2-only, and FLAIR-only modalities. Each model was subsequently tested on the complete multimodal test set to assess its generalizability. The results showcased in figure 9 clearly demonstrate that models trained on a single modality perform significantly worse, with accuracies of 73.81% (T1-trained), 70.06% (T2-trained), and 77.19% (FLAIR-trained), compared to 97.08% achieved by the proposed MNMS model. Among the single modalities, FLAIR yielded the highest performance due to its strong contrast in capturing edema and lesion regions. However, no single modality alone was sufficient to capture the full complexity of tumor structures. In contrast, the MNMS model, trained on all three modalities, effectively leveraged the complementary information from T1, T2, and FLAIR scans to achieve superior segmentation performance. This demonstrates the critical importance of multimodal fusion in medical image segmentation, as it enables the model to capture richer spatial and contextual features, making it more robust.

VI. CONCLUSION

We have proposed a multimodal brain tumor image segmentation approach on the BraTS2020 dataset with T1, T2, and FLAIR MRI modalities. This framework holds potential for integration into clinical decision-support systems, radiology workflows, and tumor progression monitoring tools. It can also support surgical planning and post-treatment assessment by providing consistent and accurate segmentation outputs. With a novel feature extraction strategy, we supplemented window-masked and noise-masked images to enhance feature learning, offering robust representation for multiple modalities. The extracted features were then fed into dedicated modality-specific encoders followed by fusion through shared encoders, enabling an exhaustive fusion of multimodal information. The final brain tumor segmentation was achieved by MNMS, which effectively captured spatial and contextual information for accurate segmentation results. Our approach demonstrated the power of modality-specific encoding coupled with shared feature fusion for brain tumor image segmentation. Leveraging multimodal MRI data enhanced the segmentation performance by taking advantage of complementary information across modalities. Future research will be directed towards architecture pipeline optimization and self-supervised learning strategies to further increase segmentation accuracy and generalization to various medical imaging datasets. In future work, we aim to extend the application of MNMS to real-time clinical datasets to further assess its adaptability and robustness in diverse clinical environments. The BraTS2020 dataset, which we currently use, is highly reliable - featuring multi-institutional pre-operative MRI scans with ground truth annotations created by board-certified neuroradiologists and validated by an expert panel. Further we plan to incorporate both this benchmark dataset and real-time clinical data to further improve our model's performance and generalization. We also plan to collaborate with medical professionals for expert validation of segmentation outputs, ensuring the model aligns with clinical standards and decision-making needs. Additionally, we will explore pipeline optimization for faster inference and improved efficiency, and adopt self-supervised learning strategies to reduce reliance on large annotated datasets. These steps will help us move closer to making the model practical, reliable, and useful in real-world clinical scenarios.

REFERENCES

[1] G. Wang, W. Li, S. Ourselin, and T. Vercauteren, "Automatic brain tumor segmentation using cascaded anisotropic convolutional neural networks," in *Proc. Int. MICCAI Brainlesion Workshop*, Sep. 2017, pp. 178–190.

[2] M. Havaei, A. Davy, D. Warde-Farley, A. Biard, A. Courville, Y. Bengio, C. Pal, P.-M. Jodoin, and H. Larochelle, "Brain tumor segmentation with deep neural networks," *Med. Image Anal.*, vol. 35, pp. 18–31, Jan. 2017.

[3] W. Wang, C. Chen, M. Ding, H. Yu, S. Zha, and J. Li, "Transbts: Multimodal brain tumor segmentation using transformer, medical image computing and computer assisted intervention-miccai 2021," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2021, pp. 109–119.

[4] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 4th Int. Conf. 3D Vis. (3DV)*, Oct. 2016, pp. 565–571.

[5] T. Koker, F. Mireshghallah, T. Titcombe, and G. Kaissis, "U-noise: Learnable noise masks for interpretable image segmentation," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2021, pp. 394–398.

[6] H. Wang, S. Xie, L. Lin, Y. Iwamoto, X.-H. Han, Y.-W. Chen, and R. Tong, "Mixed transformer U-Net for medical image segmentation," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2022, pp. 2390–2394.

[7] S. Javed, T. M. Khan, A. Qayyum, H. Alinejad-Rokny, A. Sowmya, and I. Razzak, "Advancing medical image segmentation with mini-net: A lightweight solution tailored for efficient segmentation of medical images," 2024, *arXiv:2405.17520*.

[8] M. Y. Ansari, Y. Yang, S. Balakrishnan, J. Abinahed, A. Al-Ansari, M. Warfa, O. Almokdad, A. Barah, A. Omer, A. V. Singh, P. K. Meher, J. Bhadra, O. Halabi, M. F. Azampour, N. Navab, T. Wendler, and S. P. Dakua, "A lightweight neural network with multiscale feature enhancement for liver CT segmentation," *Sci. Rep.*, vol. 12, no. 1, p. 14153, Aug. 2022.

[9] J.-B. Cordonnier, A. Loukas, and M. Jaggi, "Multi-head attention: Collaborate instead of concatenate," 2020, *arXiv:2006.16362*.

[10] H. Yu, L. Gao, H. Yu, and A. Zhang, "Vision transformer based UNet with multi-head attention for medical image segmentation," in *Proc. 36th Chin. Control Decis. Conf. (CCDC)*, May 2024, pp. 1737–1741.

[11] S. Srinivasan, K. Durairaju, K. Deeba, S. K. Mathivanan, P. Karthikeyan, and M. A. Shah, "Multimodal biomedical image segmentation using multi-dimensional U-convolutional neural network," *BMC Med. Imag.*, vol. 24, no. 1, p. 38, Feb. 2024.

[12] J. Dhar, N. Zaidi, M. Haghighat, P. Goyal, S. Roy, A. Alavi, and V. Kumar, "Multimodal fusion learning with dual attention for medical imaging," 2024, *arXiv:2412.01248*.

[13] M. Y. Ansari, I. A. Changaai Mangalote, P. K. Meher, O. Aboumarzouk, A. Al-Ansari, O. Halabi, and S. P. Dakua, "Advancements in deep learning for B-mode ultrasound segmentation: A comprehensive review," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 8, no. 3, pp. 2126–2149, Jun. 2024.

[14] M. Y. Ansari, M. Qaraqe, R. Righetti, E. Serpedin, and K. Qaraqe, "Unveiling the future of breast cancer assessment: A critical review on generative adversarial networks in elastography ultrasound," *Frontiers Oncol.*, vol. 13, Dec. 2023, Art. no. 1282536.

[15] S. Dörterler, H. Dumlu, D. Özdemir, and H. Temurtaş, "Hybridization of meta-heuristic algorithms with k-means for clustering analysis: Case of medical datasets," *Gazi Mühendislik Bilimleri Dergisi*, vol. 10, pp. 1–11, Apr. 2024.

[16] J. Xu and S. Stirenko, "Denoising self-distillation masked autoencoder for self-supervised learning," *Int. J. Image, Graph. Signal Process.*, vol. 15, no. 5, pp. 29–38, Oct. 2023, doi: 10.5815/ijigsp.2023.05.03.

[17] Y. Xie, L. Gu, T. Harada, J. Zhang, Y. Xia, and Q. Wu, "Rethinking masked image modelling for medical image representation," *Med. Image Anal.*, vol. 98, Dec. 2024, Art. no. 103304.

[18] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Proc. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.

[19] F. Wang, R. Jiang, L. Zheng, C. Meng, and B. Biswal, "3D U-Net based brain tumor segmentation and survival days prediction," in *Proc. Int. MICCAI Brainlesion Workshop*, 2019, pp. 131–141.

[20] Z. Zhao, J. Hu, Z. Zeng, X. Yang, P. Qian, B. Veeravalli, and C. Guan, "MMGL: Multi-scale multi-view global-local contrastive learning for semi-supervised cardiac image segmentation," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2022, pp. 401–405.

[21] Y. Zhou, X. Kang, F. Ren, S. Nakagawa, and X. Shan, "DEU-net: Dual encoder U-Net for 3D medical image segmentation," in *Proc. IEEE 22nd Int. Conf. Trust, Secur. Privacy Comput. Commun. (TrustCom)*, Nov. 2023, pp. 2735–2741.

[22] Z. Huang, L. Wang, and L. Xu, "DRA-net: Medical image segmentation based on adaptive feature extraction and region-level information fusion," *Sci. Rep.*, vol. 14, no. 1, p. 9714, Apr. 2024.

[23] S. Perera, P. Navard, and A. Yilmaz, "SegFormer3D: An efficient transformer for 3D medical image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 4981–4988.

[24] H. Wang, Y. Chen, C. Ma, J. Avery, L. Hull, and G. Carneiro, "Multi-modal learning with missing modality via shared-specific feature modelling," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 15878–15887.

[25] G. Xu, S. Wang, T. Lukasiewicz, and Z. Xu, "Adaptive-masking policy with deep reinforcement learning for self-supervised medical image segmentation," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2023, pp. 2285–2290.

[26] X. Wang, R. Wang, B. Tian, J. Zhang, S. Zhang, J. Chen, T. Lukasiewicz, and Z. Xu, "MPS-AMS: Masked patches selection and adaptive masking strategy based self-supervised medical image segmentation," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2023, pp. 1–5.

[27] C. M. I. Afsa, M. Y. Ansari, S. Paul, O. Halabi, E. Alataresh, J. Shah, A. Hamze, O. Aboumarzouk, A. Al-Ansari, and S. P. Dakua, "Development and validation of a class imbalance-resilient cardiac arrest prediction framework incorporating multiscale aggregation, ICA and explainability," *IEEE Trans. Biomed. Eng.*, vol. 72, no. 5, pp. 1674–1687, May 2025.

[28] M. Y. Ansari, M. Qaraqe, R. Righetti, E. Serpedin, and K. Qaraqe, "Enhancing ECG-based heart age: Impact of acquisition parameters and generalization strategies for varying signal morphologies and corruptions," *Frontiers Cardiovascular Med.*, vol. 11, Jul. 2024, Art. no. 1424585.

[29] M. Yaqoob, M. Y. Ansari, M. Ishaq, U. Ashraf, S. Pavuluri, A. Rabbani, H. S. Rabbani, and T. D. Seers, "FluidNet-lite: Lightweight convolutional neural network for pore-scale modeling of multiphase flow in heterogeneous porous media," *Adv. Water Resources*, vol. 200, Jun. 2025, Art. no. 104952.

[30] M. A. Dursun and P. Ö. Kavas, "Beyond diagnosis: Cross-dataset evaluation of risk factors for thyroid cancer recurrence," *Artif. Intell. Stud.*, vol. 8, no. 1, pp. 38–69, Jun. 2025.

[31] A. Hatamizadeh, Y. Tang, V. Nath, D. Yang, A. Myronenko, B. Landman, H. R. Roth, and D. Xu, "UNETR: Transformers for 3D medical image segmentation," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2022, pp. 1748–1758.

[32] T. Henry, A. Carré, M. Lerousseau, T. Estienne, C. Robert, N. Paragios, and E. Deutsch, "Brain tumor segmentation with self-ensembled, deeply-supervised 3D U-net neural networks: A BraTS 2020 challenge solution," in *Proc. Int. MICCAI Brainlesion Workshop*, 2020, pp. 327–339.

K. HEMALATHA received the M.E. degree in computer science and engineering from Anna University, Chennai, India, in 2019, and the Ph.D. degree from the College of Engineering, Guindy, Anna University. She is currently an Assistant Professor with Vellore Institute of Technology, Chennai. Her research interests include medical image processing and computer vision.

P. R. VISHNU VARDHAN is currently pursuing the Bachelor of Technology (B.Tech.) degree in computer science and engineering with a specialization in artificial intelligence and robotics with Vellore Institute of Technology, Chennai. He is passionate about deep learning and neural networks and aspires to contribute to advancements in these fields.

ALFRED DHARMARAJ ARAVINDRAJ is currently pursuing the Bachelor of Technology (B.Tech.) degree in computer science and engineering with a specialization in artificial intelligence and robotics with Vellore Institute of Technology, Chennai. His research interests include deep learning, computer vision, and autonomous systems. He aims to contribute to advancements in artificial intelligence and deep learning.

S. HARI HARA SUDHAN is currently pursuing the Bachelor of Technology (B.Tech.) degree in computer science and engineering with a specialization in cyber-physical systems with Vellore Institute of Technology, Chennai. He is passionate about deep learning, particularly in representation learning and intelligent data processing, and aspires to contribute to advancements in medical imaging and healthcare automation.